Lecture 23

Tuesday, 29 September 2026 8:23 AM

Recap: $\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} S_n$, $S_n = a_1 + a_2 + \cdots + a_n$

$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ for $-1 < x < 1$

(Necessary Cond.) $\sum_{n=p}^{\infty} a_n$ Cgs then $\lim_{n\to\infty} a_n = 0$

P

$P \Rightarrow Q$

Converse $Q \Rightarrow P$

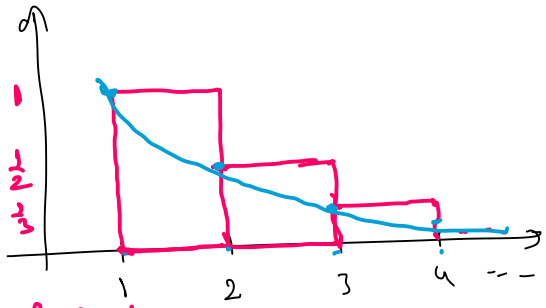
Converse: If $\lim_{n\to\infty} a_n = 0$ then $\sum_{n=1}^{\infty} a_n$ Cgs ?

"need not be true"

Eg: $\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots$

$a_n = \frac{1}{n} \longrightarrow 0$ as $n \to \infty$

But $\sum_{n=1}^{\infty} \frac{1}{n}$ dgs.



$f(x) = \frac{1}{x}$

$1 \le x < \infty$

Sum of areas of rectangles:

$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots = \sum_{n=1}^{\infty} \frac{1}{n} >$

Area under $f(x) = \frac{1}{x}$ $1 \le x < \infty$

$$\int_{x=1}^{\infty} \frac{1}{x}\,dx = \lim_{A\to\infty} \int_{x=1}^{A} \frac{1}{x}\,dx = \lim_{A\to\infty} \ln x \Big|_{x=1}^{A}$$

$= \lim_{A\to\infty} \ln A - \ln 1$

$= \infty$ Not a

Area > Area

finite num

$\therefore \sum_{n=1}^{\infty} \frac{1}{n}$ dgs.

Tests for cgs of $\sum_{n=1}^{\infty} a_n$:

Comparison Test:

If $a_n \geq 0$, $b_n \geq 0$ $\forall n \in \mathbb{N}$ and

if $a_n \geq c\, b_n$ $\forall n \geq N_0$

Then (i) $\sum a_n$ cgs $\Longrightarrow$ $\sum b_n$ cgs

(ii) $\sum b_n$ dgs $\Longrightarrow$ $\sum a_n$ dgs

Eg1): Determine the cgs/dgs of $\sum_{n=2}^{\infty} \frac{1}{\log n}$

$\frac{1}{\log 2} + \frac{1}{\log 3} + \frac{1}{\log 4} + \cdots = ?$

$a_n = \frac{1}{\log n} \longrightarrow 0$ as $n \to \infty$

(Necessary cond)

Consider $b_n = \frac{1}{n}$

Recall: $\log n \leq n$ $\forall n \geq 2$

$\Longrightarrow \frac{1}{\log n} \geq \frac{1}{n}$ $\forall n \geq 2$

($\frac{1}{\log n} = a_n$, $\frac{1}{n} = b_n$)

$\sum a_n = \sum \frac{1}{\log n}$ dgs ?

WKT, $\sum b_n = \sum \frac{1}{n}$ dgs

$\uparrow$ Comp. Test.

Eg2): $\sum \frac{1}{(\log n)^2}$

$n = 2$

$a_n = \frac{1}{(\log n)^2} \longrightarrow 0$ as $n \to \infty$ ✓

$b_n = \frac{1}{n}$

$(\log n)^2 \leq n$

$\forall n \geq$ □

$\lim_{x \to +\infty} \frac{(\log n)^a}{n^b} = 0$

$a, b > 0$

$\frac{1}{(\log n)^2} \geqslant \frac{1}{n}$

$\sum \frac{1}{n}$ dgs $\Longrightarrow \sum \frac{1}{(\log n)^2}$ dgs

Ex: Find all values of $s \in \mathbb{R}$ for which $\sum_{n=2}^{\infty} \frac{1}{(\log n)^s}$ cgs / dgs.

Root Test:

Let $a_n \geq 0$ $\forall n \in \mathbb{N}$, $L = \lim_{n \to \infty} (a_n)^{1/n}$

Then $\sum_{n=1}^{\infty} a_n$

(i) Cgs if $L < 1$

(ii) Dgs if $L > 1$

(iii) Inconclusive if $L = 1$

Eg 1): $\sum_{n=1}^{\infty} (n^{1/n} - 1)^n$. Determine the cgs of this series.

$a_n = (n^{1/n} - 1)^n \longrightarrow 0$ as $n \to \infty$ ✓

$L = \lim_{n \to \infty} a_n^{1/n}$

$= \lim_{n \to \infty} (n^{1/n} - 1) = 0 < 1$

$0 < n^{1/n} - 1 < n^{1/n}$ cgs

$\therefore$ By Root Test, $\sum_{n=1}^{\infty}$ (illegible)

Eg2): $\sum_{n=2}^{\infty} \frac{1}{(\log n)^n}$ Cgs

Ratio Test: Let $a_n > 0 \quad \forall n \in \mathbb{N}$

and let $L = \lim_{n \to \infty} \frac{a_{n+1}}{a_n}$. Then $\sum_{n=1}^{\infty} a_n$

(i) Cgs if $L < 1$

(ii) Dgs if $L > 1$

(iii) Inconclusive if $L = 1$.

Eg1): Check the Cgs/ dgs of $\sum_{n=1}^{\infty} \frac{1}{n!}$

$a_n = \frac{1}{n!} \longrightarrow 0$ as $n \to \infty$ ✓

$$L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{\frac{1}{(n+1)!}}{\frac{1}{n!}}$$

$$= \lim_{n\to\infty} \frac{1}{n+1} = 0 < 1$$

$\therefore$ By Ratio Test, $\sum_{n=1}^{\infty} \frac{1}{n!}$ Cgs

$\hookrightarrow e - 1$

$\sum a_n$

$\lim_{n\to\infty} a_n$

$= 0$ → Comp. Test, Ratio Test, Root Test, Telescoping series

$\neq 0$ → Dgs

Ex(i) $\sum_{n=1}^{\infty} \frac{1+\sqrt{n}}{(n+1)^3 - 1}$

Ex: $\sum_{n=1}^{\infty} n^3 \left(\sqrt{2} + (-1)^n\right)^n$

(2) $\sum_{n=1}^{\infty} \frac{1}{3^n}$